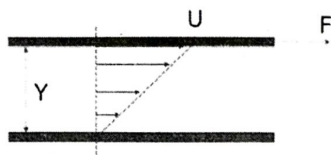


ChE 311 Quiz 1 Spring 2019

Closed book and closed notes
5 minutes, 5 points

3

1. A type of gear oil was filled between two parallel plate with a gap distance Y , the top plate was moving at constant speed U in x direction under a constant force F . The bottom plate was stationary. Initially, the viscosity of the gear oil was μ . Overtime, the gear oil viscosity was reduced to 0.5μ . Which of the following statement(s) is (are) true.



$$\tau_{yx} = \frac{F_A}{A} = \mu \frac{\Delta u_x}{\Delta y}$$

$$\mu \frac{\Delta u_x}{\Delta y} = \frac{1}{2} \mu \frac{\Delta u_x}{\Delta y}$$

$$F_A = A \mu \frac{\Delta u_x}{\Delta y}$$

$$F_A = A \frac{1}{2} \mu \frac{\Delta u_x}{\Delta y}$$

- ☒ A. With the same Y and F , the top plate moved at $2U$.
☒ B. With the same Y and F , the top plate moved at $\frac{1}{2} U$.
☒ C. With the same Y and U , the force is reduced to $\frac{1}{2} F$.
☒ D. None of above.

Answer: C

2. U and V are velocity vectors.

2.1 $\nabla \cdot V$ is a C.

- (A) vector (B) tensor (C) scalar (D) not enough information to decide

2.2 ∇V is a A.

- (A) vector (B) tensor (C) scalar (D) not enough information to decide

2.3 $\nabla \cdot VV$ is a A.

- (A) vector (B) tensor (C) scalar (D) not enough information to decide

2.4 UV is the same as VU .

- (A) True (B) False